

Technology Comparison for ProofRoad

Primary project source: ProofRoad v2, SIH 2026 PS 26007, uploaded by the user. The document describes ProofRoad as an engineering proposal and explicitly says it is not a safety certification. Page references below refer to the uploaded ProofRoad v2 document.

1. Perception technologies

Technology	Primary strength	Dense-fog behavior	ProofRoad role
77 GHz radar	Range + speed	Least affected among selected sensors	Primary clear-road proof
60 GHz radar	Accessible evaluation/demo hardware	Good prototype modality; not a mine-grade deployment claim	Demo/stand-in
RGB	Object naming, road edges	Near-blind at 3–5 m visibility	Assist only
Thermal	Warm-object detection	Can work better than visible light in thin mist/dust; dense fog still limits range	Shorten-only
NIR	Dusk/night/dust contrast	Scatters in dense fog similar to visible light	Shorten-only cross-check
LiDAR	3D geometry	ProofRoad states dense fog degrades it strongly	Not selected

2. Communication technologies

Link	ProofRoad use	Design reason
ESP-NOW	Fast truck beacons, roughly 10 messages/s in proposal	Low-latency local peer messaging
LoRa	Fog posts, emergency beacon, long-range relay	Slow/long-range traffic only
Central broker	Not in safety path	Avoid single control-room dependency

3. Positioning

GPS is optional in the ProofRoad design. Position along a known route can be supported with checkpoints, IMU and wheel-distance cross-checks; increasing position uncertainty shortens the proven distance (p. 14).

For production, the field design should specify the required positioning accuracy for the particular haul road, especially near curves and merge points.

4. Compute architecture

Safety MCU: radar input, corridor calculation, alert logic, watchdog and local outputs.

Assist computer: RGB, thermal and NIR processing, object naming, fog checks and dashboard integration.

This separation limits the consequence of AI/software-computer failure and makes the core safety calculation deterministic and replayable.